Exam Date & Time: 25-May-2023 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - MAY/JUNE 2023
SUBJECT: BIO 1071-CHM/ BIO 1051-B/ BI0 1071-B - BIOLOGY FOR ENGINEERS

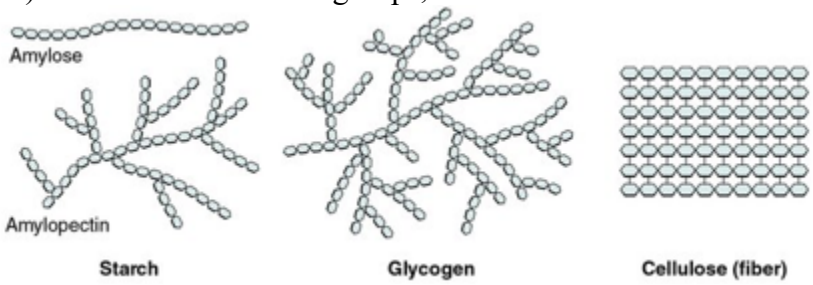
BIOLOGY FOR ENGINEERS [BIO 1071-CHM]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

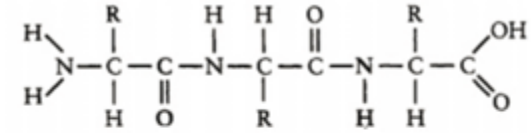
1A) Given below is an illustration of polysaccharides. Analyse the structure and answer the following questions.
i) What is the reason for diverse structure, eventhough monomers are sugar molecules?
ii) Why cellulose is fiber like, but amylose is a straight chain?
iii) All of them have -OH groups, but cellulose is insoluble in water. Justify.

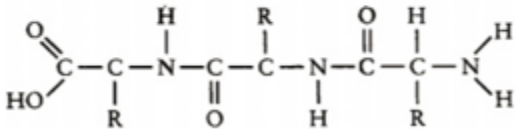


(3)

(1+1+1 = 3 marks)

1B) Here you will get two proteins chains.





(3)

i) Illustrate the way in which they interact and what will be the nature of interaction?
ii) In this which group plays NO part in the protein secondary structure? [Copy the figure and highlight]Why?
(2+1 = 3 marks)

1C) In alien text books the life on earth may be best described as "carbon nanofibers". What makes carbon that forms the basis of earth life? (4)

2A) A typical Mendelian cross starts with a tall pea plant bearing round seeds and a dwarf pea plant bearing wrinkled seeds (3)

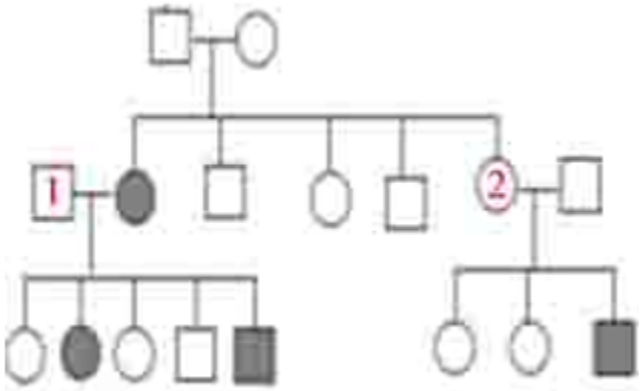
1. Construct a Punnett square explaining the F2 generation genotypes.
2. F2 generation produced 19280 seeds. (2a) Predict the number of true breeding recessive phenotypes (2b) Calculate the number of individuals homozygous for one character and heterozygous for the other character.
(1+2 = 3 marks)

2B) You are a teaching assistant for Biology for Engineers. You have pure-breeding wheat plants of a particular kind which produce long and thin seeds (Dominant, LLTT). You also have pure-breeding plants of short and thick seeds (ttll). You asked your student project assistant to cross the two and to give you the results of F1. Unfortunately, your student forgets to record the phenotype of the F1 plants in her notebook, though she does remember that they all looked the same. The F1 was test crossed and the results are:

(3)

Long wheat, thick seeds = 186
Long wheat, thin seeds = 564
Short wheat, thick seeds = 63
Short wheat, thin seeds = 187
1. Construct a Punnett square explaining the test cross.
2. Calculate the recombination frequency.
3. What is your conclusion from the results of test cross?
(1+1+1 = 3 mark)

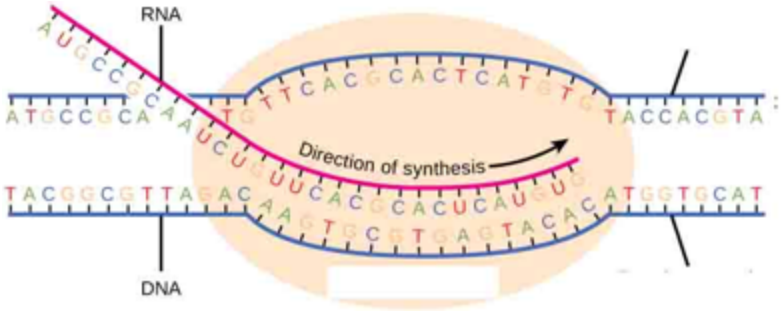
2C) Given below is a pedigree chart. Analyze and answer the following questions.



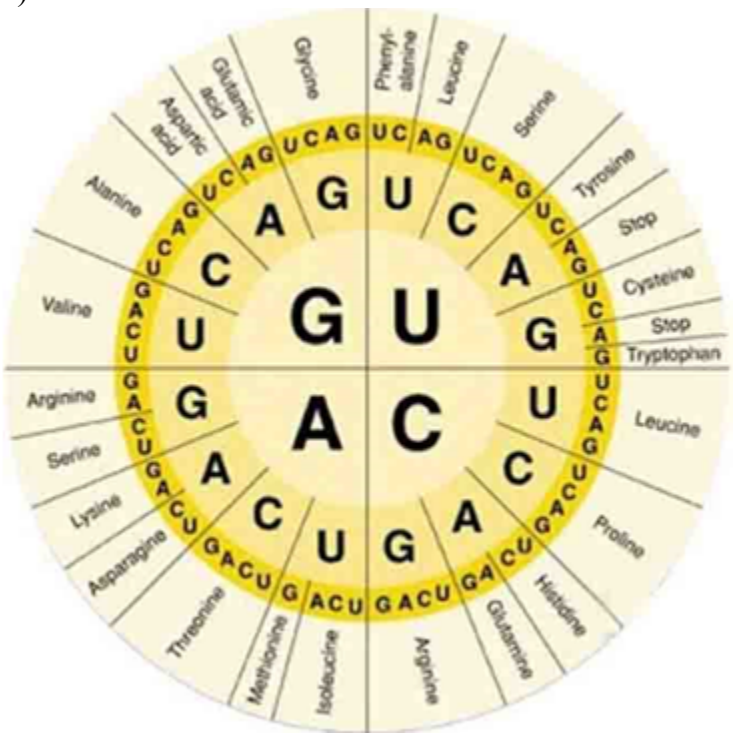
(4)

i) What is the best possible mode of inheritance and why?
ii) Predict the genotype of the individual 1 and individual 2.
iii) For the individual 2, calculate the probability of the fourth child being unaffected phenotype?
(2+1+1 = 4 marks)

3A) An illustration of mRNA synthesis is depicted below (3)



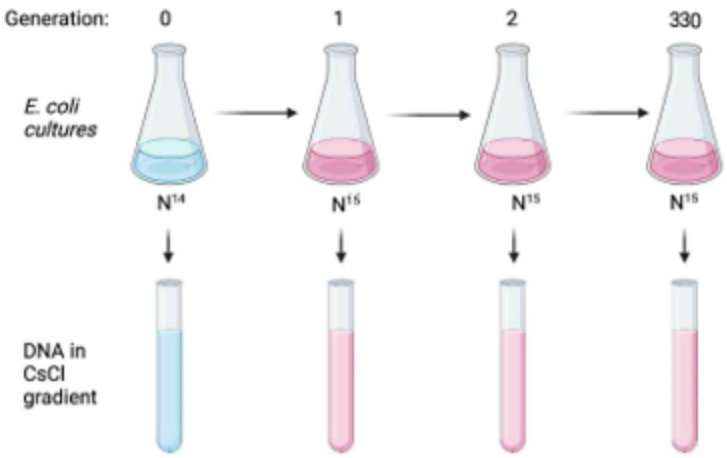
i) Copy the figure and label the ends of mRNA and DNA.
ii) Find the template strand and non-template strand as well as the enzyme required for this process.
iii) Find the anticodon for the code CCG in the mRNA and find the amino acid it codes.



(1+1+1 = 3 marks)

3B) An engineer is always trying to improve an existing system. We call living systems as best engineers. What are the ways in which living systems can improves by itself? Justify by giving some relevant example. (3)

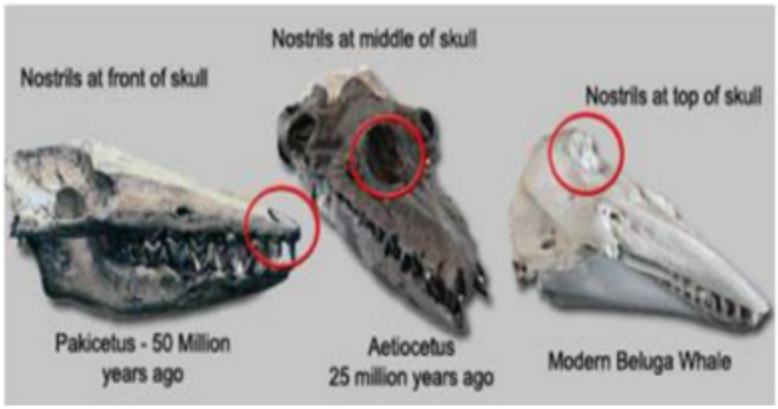
3C) Given below is a illustration of Meselson and Stahl experiment for a synthetic bacteria in which the mode of replication is **CONSERVATIVE**. The student started with **N^{14} cultures and then shifted to N^{15}**. (4)



i) Copy the figure and mark the band accumulation in each of the generation given.
ii) Calculate the percentage of DNA at the end of 2nd generation bearing N^{14} N^{15} strand.
iii) Why $CsCl_2$ is use as a medium for centrifugation here? Justify.
(2+1+1 = 4 marks)

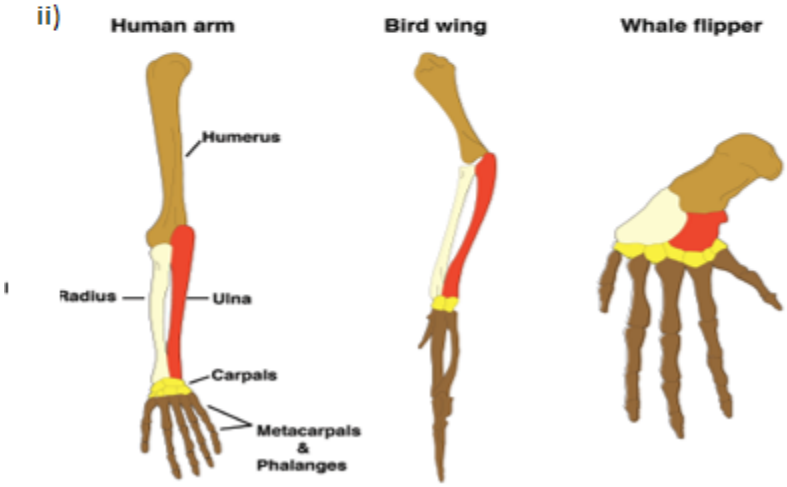
4A) Various evidence has been found to prove the evolutionary development in living organisms. By looking at the given figure, what type of evidence can be explained based on the difference and development of the organs.

i)



(3)

ii)



4B) What is the concept of interaction in the following cases? Explain. (3)

Example	Concept	Explanation

4C) The theory of use and disuse was propounded by one of the scientists who provided the information about the classification of invertebrates in his book, and he is the one who coined the term invertebrates.
i) How did he explain it? (4)
ii) How the theory was later modified? Explain logically
(2+2 = 4 marks)

5A) "Doing it nature's way" has the potential to change the way we grow food, make materials, harness energy, heal ourselves, store information, and conduct business. In each case, nature would be model, measure, and mentor. As an engineer how do you perceive these ideas explain with examples? (3)

5B) i) Given below is the diagram of Galapagos Shark. It repels bacteria from its skin. Develop a bioinspiration idea/mechanism from it. (3)

ii) Understand and analyze the following figure from engineering perspective. How does

nature strengthen with minimal material and how did engineers can bioinspire from it? Give examples of the inspirations.

iii) How can we get inspired from the organisms in the following image. How do we improve our existing problems referring to this example.

(1+1+1 = 3 marks)

5C) In 1796, Edward Jenner, an English physician noticed that dairy workers who had caught relatively mild disease cowpox from cows seemed to be immune to smallpox. He found that rubbing pus from cowpox sores into scratches in the skin prevented people from coming down with smallpox later. Almost a century later, Louis Pasteur found a safer way to prepare vaccines. He discovered how to attenuate (kill them so that they will infect, but no disease is caused) microorganisms until they could no longer cause disease. Smallpox was not only the first disease to be prevented by vaccination but also the first disease to be officially eradicated by human efforts. (4)

i) How does vaccination protect the body from getting disease?
ii) Is it safe to use attenuated microorganisms as vaccines?
iii) Explain why rubbing pus from cowpox sores into scratches in the skin prevented people from getting smallpox.
iv) Give the reason why several diseases such as common cold, chicken pox, corona etc have remained ineffective or partially effective against vaccines.
(1+1+1+1 = 4 marks)

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